

Design and Analysis of Algorithms

Date: Nov 25, 2025

Course Instructor(s)

Dr. Fahad Sherwani

Quiz - 2

Total Time: 50 Min

Total Marks: 2.5

Total Questions: 3

**** Plagiarism of any kind is highly discouraged and will result in Zero marks ****

Roll No

Section

Student Signature

Do not write below this line

Attempt all the questions.

CLO #4: To construct and analyze real world problems solutions using different algorithms design techniques i.e. Brute Force, Divide and Conquer, Dynamic Programming, Greedy Algorithms.

Q1: A telecom provider wants to compute the **least-cost routing path** between two network towers in its backbone network. The network is modeled as a **directed weighted graph** $G = (V, E)$, where:

- Each vertex represents a **tower**.
- Each directed edge (u, v) has a **cost** $w(u, v)$ representing the routing cost from tower u to tower v .
- Some links can have **negative costs** (e.g., due to discounts or special contracts), Consider the following network of **4 towers** (1 to 4):
 - Edge $(1 \rightarrow 2)$ with cost **3**
 - Edge $(1 \rightarrow 3)$ with cost **8**
 - Edge $(2 \rightarrow 3)$ with cost **-2**
 - Edge $(2 \rightarrow 4)$ with cost **5**
 - Edge $(3 \rightarrow 4)$ with cost **1**

- a) Explain why Dynamic Programming is an appropriate technique to solve the least-cost path problem in such a network instead of relying on a purely recursive approach.
- b) Design a **Dynamic Programming algorithm** to compute the minimum-cost route from any **source** tower to any **destination** tower in a directed graph. Also dry run your approach using the values above.

CLO #4: To construct and analyze real world problems solutions using different algorithms design techniques i.e. Brute Force, Divide and Conquer, Dynamic Programming, Greedy Algorithms.

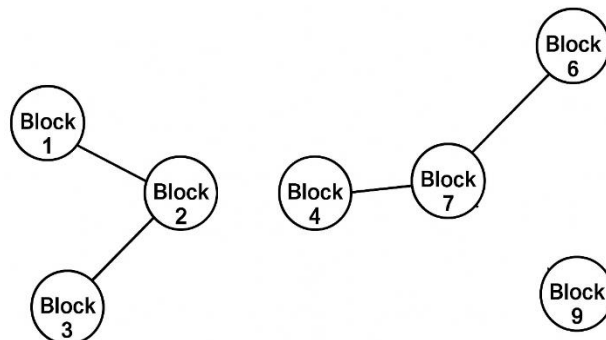
Q2: A city planning department is analyzing the layout of several neighborhoods to understand which areas are internally reachable by foot using the available pedestrian paths.

National University of Computer and Emerging Sciences

Karachi Campus

The map is represented as an undirected graph, where:

Each node represents a neighborhood block, and Each edge represents a walkable path connecting two blocks.



- Design an algorithm that identifies the blocks that are connected to each other and returns them as connected neighborhood groups within a map.
- Dry run your approach over the sample map given above.

CLO #4: To construct and analyze real world problems solutions using different algorithms design techniques i.e. Brute Force, Divide and Conquer, Dynamic Programming, Greedy Algorithms.

Q3: A manufacturing company produces customized electronic devices. The production of each device requires a set of tasks, and many of these tasks have dependencies — certain tasks must be completed before others can begin. The engineering team provides the following partial list of tasks and their precedence requirements:

Task	Description	Must be done before
A	Assemble the circuit board	C, D
B	Print the casing	E
C	Install microcontroller	F
D	Install sensors	F
E	Paint and finish the casing	G
F	Integrate hardware modules	G
G	Final device assembly	—

- Model the above production process as a directed acyclic graph (DAG) where each task is a vertex and each dependency is a directed edge.
- Design an algorithm to produce a valid manufacturing order in which tasks can be completed without violating dependencies. Based on the task list above, provide one valid ordering of the tasks as well.